NAME: VAISHNAVI E ROLL NO: 231901059

EX. NO. 5

INTEGRATION WITH EXTERNAL SYSTEMS USING HYPERLEDGER FABRIC AIM

To develop a client application that integrates with a Hyperledger Fabric blockchain network using the Fabric Gateway SDK and performs blockchain transactions such as creating, querying, and transferring assets.

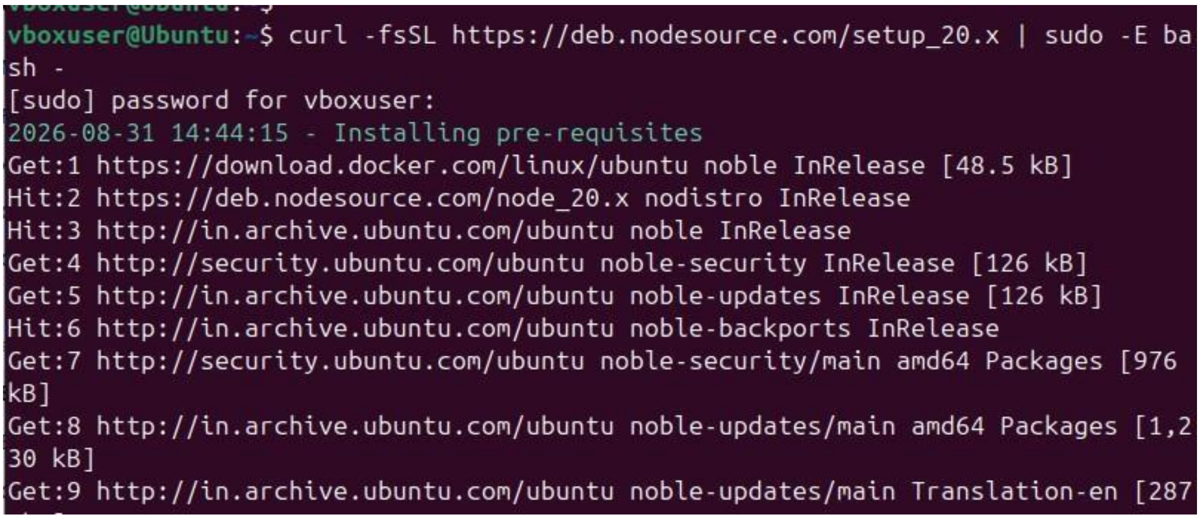
SOFTWARE REQUIREMENTS

- *Kali Linux / Ubuntu*
- *Docker*
- *Docker Compose*
- *Node.js Version 20 or later*
- *npm*
- *Git*
- *Hyperledger Fabric Samples*
- *Hyperledger Fabric Binaries*
- *Fabric Gateway SDK (Node.js)*
- *Visual Studio Code*

PROCEDURE

Step 1: Upgrade Node.js to Version 20+

Install Node.js version 20 or later using NodeSource. curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash –



```
vboxuser@Ubuntu:~$ curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E ba
sh -
[sudo] password for vboxuser:
2026-08-31 14:44:15 - Installing pre-requisites
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Hit:2 https://deb.nodesource.com/node_20.x nodistro InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [976
kB]
Get:8 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,2
30 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [287
```

Install Node.js.

sudo apt install -y nodejs Check

the installed Node.js version.

node -v

Ensure that the installed version is v20.x.x or later before continuing.

```
vboxuser@Ubuntu:~$ sudo apt install -y nodejs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nodejs is already the newest version (20.20.2-1nodesource1).
The following packages were automatically installed and are no longer required:
  bridge-utils containerd docker.io libfwupd2 libgl1-amber-dri libglapi-mesa
  libwoff1 node-busboy node-cjs-module-lexer node-undici node-xtend pigz
  python3-compose python3-docker python3-dockerpty python3-docopt
  python3-dotenv python3-packaging python3-texttable python3-websocket runc
  ubuntu-fan
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 43 not upgraded.
vboxuser@Ubuntu:~$ node -v
v20.20.2
```

Step 2: Start the Fabric Network *Navigate to the Hyperledger*

Fabric test

network. cd ~/fabric-

dev/fabricsamples/test-network Stop any

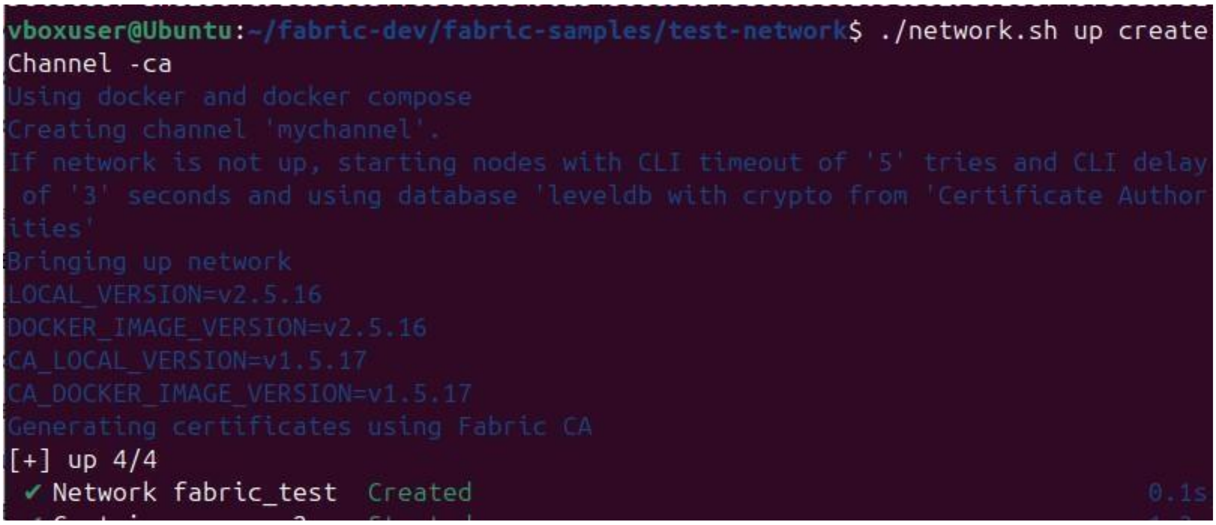
previously running Fabric network.

./network.sh down

```
v20.20.2
vboxuser@Ubuntu:~$ cd ~/fabric-dev/fabric-samples/test-network
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Using docker and docker compose
Stopping network
[+] down 10/10
 ✔ Container ca_orderer                      Removed    0.5s
 ✔ Container peer0.org2.example.com          Removed    0.5s
 ✔ Container ca_org1                         Removed    0.5s
 ✔ Container ca_org2                         Removed    0.5s
 ✔ Container orderer.example.com             Removed    0.5s
 ✔ Container peer0.org1.example.com          Removed    0.5s
 ✔ Volume compose_orderer.example.com        Removed    0.0s
 ✔ Volume compose_peer0.org1.example.com     Removed    0.1s
 ✔ Volume compose_peer0.org2.example.com     Removed    0.1s
 ✔ Network fabric_test                       Removed    0.3s
```

Start the Fabric network and create the application channel with Certificate Authority.

./network.sh up createChannel -ca



Step 3: Deploy the JavaScript Chaincode

Deploy the Asset Transfer Basic JavaScript chaincode.

./network.sh deployCC -ccl javascript -ccn basic -ccp ../assettransferbasic/chaincode-javascript

Step 4: Navigate to the Application Directory *Move to the JavaScript application directory.*

cd ../asset-transfer-basic/applicationgateway-javascript List the files in the application directory.

ls

Inspect the package.json file to identify the application entry point and available scripts. cat package.json

NAME: VAISHNAVI E ROLL NO: 231901059

Check the main field and the scripts section in the package.json file.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ cd ../asset-transfer-basic/application-gateway-javascript
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ ls
eslint.config.mjs  package.json  src
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gateway-javascript$ cat package.json
{
    "name": "asset-transfer-basic",
    "version": "1.0.0",
    "description": "Asset Transfer Basic Application implemented in JavaScript using fabric-gateway",
    "engines": {
        "node": ">=20"
    },
    "scripts": {
        "lint": "eslint src",
        "pretest": "npm run lint",
        "start": "node src/app.js"
```

Step 5: Clean Install Node.js Dependencies *Remove the existing*

node_modules directory and package lock file. rm -rf

node_modules

packagelock.json

Install the required Node.js packages. npm install

Step 6: Run the Client Application

Run the application using the available start script.

npm start

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gat
eway-javascript$ rm -rf node_modules package-lock.json
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gat
eway-javascript$ npm install
npm warn deprecated eslint@9.39.5: This version is no longer supported. Please s
ee https://eslint.org/version-support for other options.

added 122 packages, and audited 123 packages in 39s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gat
eway-javascript$ npm start

> asset-transfer-basic@1.0.0 start
> node src/app.js

channelName:       mychannel
```

If a start script is not available, run the application directly using its entry point. node src/app.js

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/asset-transfer-basic/application-gat
eway-javascript$ node src/app.js
channelName:       mychannel
chaincodeName:     basic
mspId:             Org1MSP
cryptoPath:        /home/vboxuser/fabric-dev/fabric-samples/test-network/organiz
ations/peerOrganizations/org1.example.com
keyDirectoryPath:  /home/vboxuser/fabric-dev/fabric-samples/test-network/organiz
ations/peerOrganizations/org1.example.com/users/User1@org1.example.com/msp/keyst
ore
certDirectoryPath: /home/vboxuser/fabric-dev/fabric-samples/test-network/organiz
ations/peerOrganizations/org1.example.com/users/User1@org1.example.com/msp/signc
erts
tlsCertPath:       /home/vboxuser/fabric-dev/fabric-samples/test-network/organiz
ations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.cr
t
peerEndpoint:      localhost:7051
peerHostAlias:     peer0.org1.example.com
```

The client application connects to the Hyperledger Fabric network through the Fabric Gateway SDK and performs the required blockchain operations.

Step 7: Initialize and Query the Ledger

The client application invokes the InitLedger transaction to initialize the ledger with sample assets.

It then evaluates the GetAllAssets transaction to retrieve the assets stored in the world state.

The application displays the retrieved asset information.

Step 8: Create a New Asset

The client application submits a CreateAsset transaction to the blockchain network. The transaction creates a new asset in the ledger with the required asset details.

Step 9: Transfer Asset Ownership

The application submits a TransferAsset transaction to transfer ownership of an existing asset to another owner.

The transaction updates the asset information stored in the world state.

```
*** Result: [
  {
    AppraisedValue: 300,
    Color: 'blue',
    ID: 'asset1',
    Owner: 'Tomoko',
    Size: 5,
    docType: 'asset'
  },
  {
    AppraisedValue: 1300,
    Color: 'yellow',
    ID: 'asset1788187865636',
    Owner: 'Saptha',
    Size: 5
  },
  {
    AppraisedValue: 1300,
    Color: 'yellow',
    ID: 'asset1788187974903',
    Owner: 'Saptha',
    Size: 5
```

Step 10: Query the Updated Asset

The client application evaluates the ReadAsset transaction to retrieve the updated asset information.

The returned information is displayed by the application to verify that the ownership change was successfully recorded on the blockchain.

Step 11: Stop the Fabric Network

Navigate back to the Fabric test network directory. cd ../../test-

network

Stop and remove the Fabric network.

./network.sh down

APPLICATION WORKFLOW

Start Fabric Network

NAME: VAISHNAVI E ROLL NO: 231901059

|

▼

Deploy Chaincode

|

▼

Install Node.js Packages

|

▼

Run Client Application

|

▼

Connect using Gateway SDK

|

▼

Initialize Ledger

|

▼

Submit Transaction

|

▼

Query Ledger |

▼

Create / Transfer Asset

|

▼

Query Updated Asset

|

▼

Display Asset Details

|

▼

Stop Fabric Network

RESULT

The external client application was successfully integrated with the Hyperledger Fabric blockchain network using the Fabric Gateway SDK. The application established a secure connection, initialized and queried the ledger, created and transferred assets, and retrieved updated asset information.